EES 301: Global biogeochemical cycles and Climate change

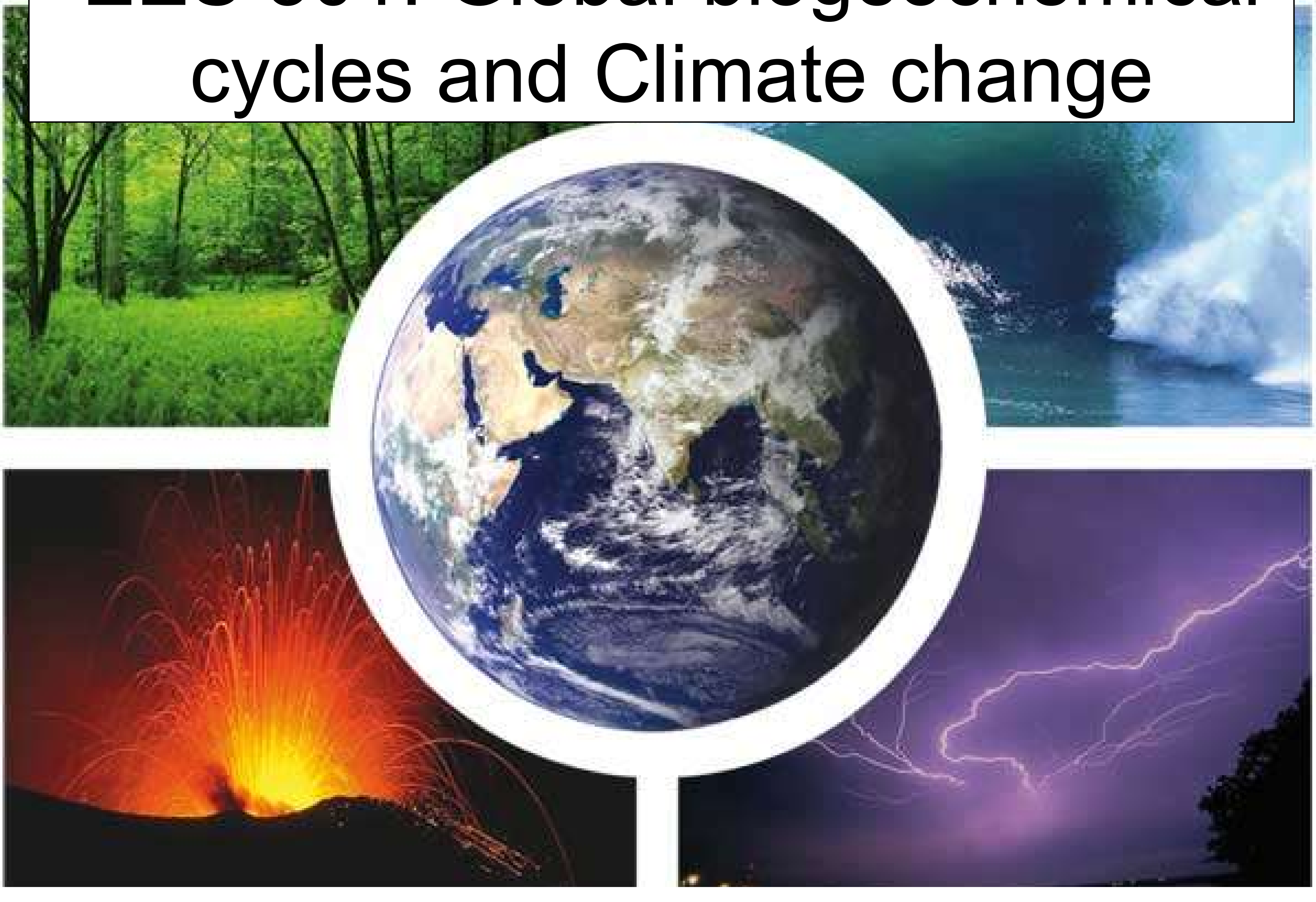


Table 7-2

Major gaseous constituents of dried air

		Average concentration (volume fraction)
N_2	Nitrogen	0.78084
O_2	Oxygen	0.20946
^{40}Ar	Argon	9.34×10^{-3}
CO_2	Carbon dioxide	3.7×10^{-4}
Ne	Neon	1.8×10^{-5}
He	Helium	5.24×10^{-6}
CH_4	Methane	1.7×10^{-6}
Kr	Krypton	1.13×10^{-6}
H_2	Hydrogen	5×10^{-7}
N_2O	Nitrous oxide	3×10^{-7}
Xe	Xenon	8.7×10^{-8}
CO	Carbon monoxide	$5–20 \times 10^{-8}$
OCS	Carbonyl sulfide	5×10^{-10}
O_3	Ozone	
	Troposphere (clean)	5×10^{-8}
	Troposphere (polluted)	4×10^{-7}
	Stratosphere	1×10^{-7} to 6×10^{-6}

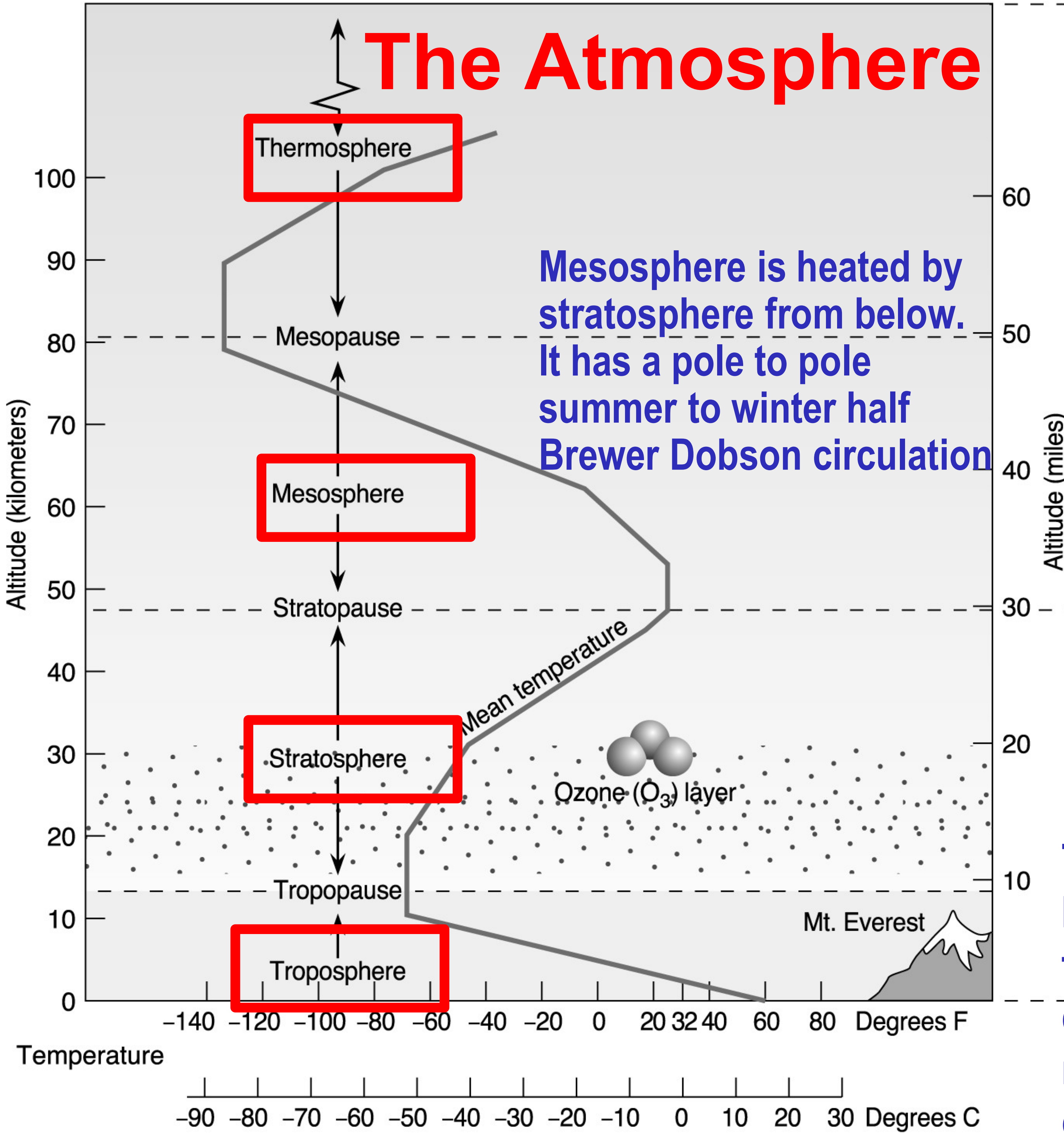
The Atmosphere

The atmosphere is a thin layer of gas uniformly covering the whole Earth. Its main constituents are nitrogen (N_2), oxygen (O_2), argon (Ar), water (H_2O) gas). The concentration of these gasses is typically given in %.

CO_2, Ne, He, Kr, CH_4, and sometimes CO are typically reported in parts per million ppm

O_3, NO, N_2O, and SO_2 are typically reported in parts per billion ppb

The Atmosphere



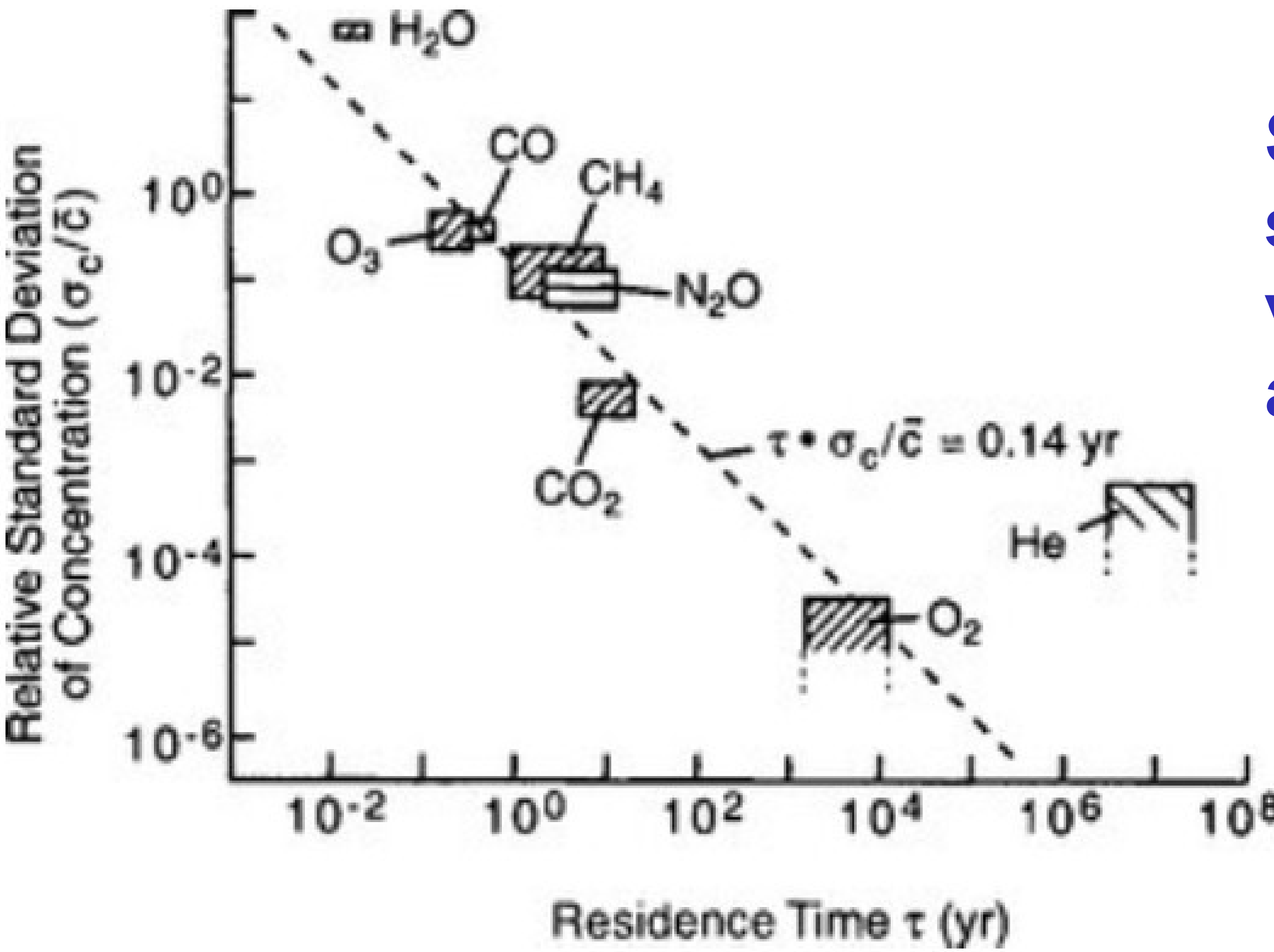
Mesosphere is heated by stratosphere from below. It has a pole to pole summer to winter half Brewer Dobson circulation

Photo dissociation and photo ionization (exothermic) by x-rays and hard UV are the heat source of the Thermosphere and greatest at the edge of the magnetosphere. A thermometer would show low T due to less air.

Photolysis of O_2 and O_3 (exothermic) is the heat source of the stratosphere and greatest at the top. Slow vertical mixing.

Troposphere is heated via black body radiation of Earth from the bottom. Temperature decreases with height resulting in rapid thermal convection

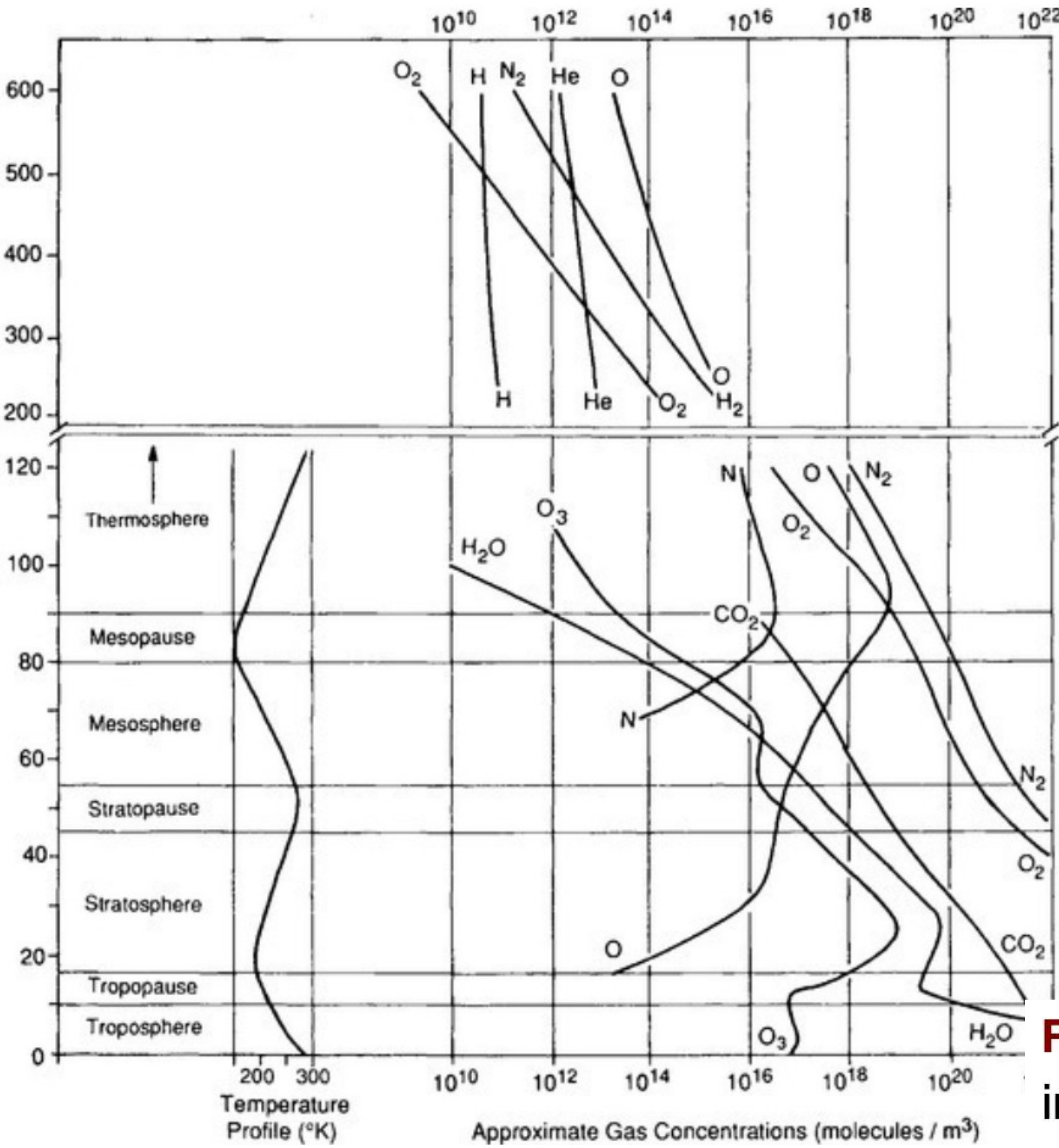
Long lived constituents are well mixed on a global scale



Shorter lived species show a high variability in space and time

FIG. 4-2 Inverse relationship between relative standard deviation of atmospheric concentration and turnover time for important trace chemicals in the troposphere. (Modified from Junge (1974) with permission from the Swedish Geophysical Society.)

The Atmosphere

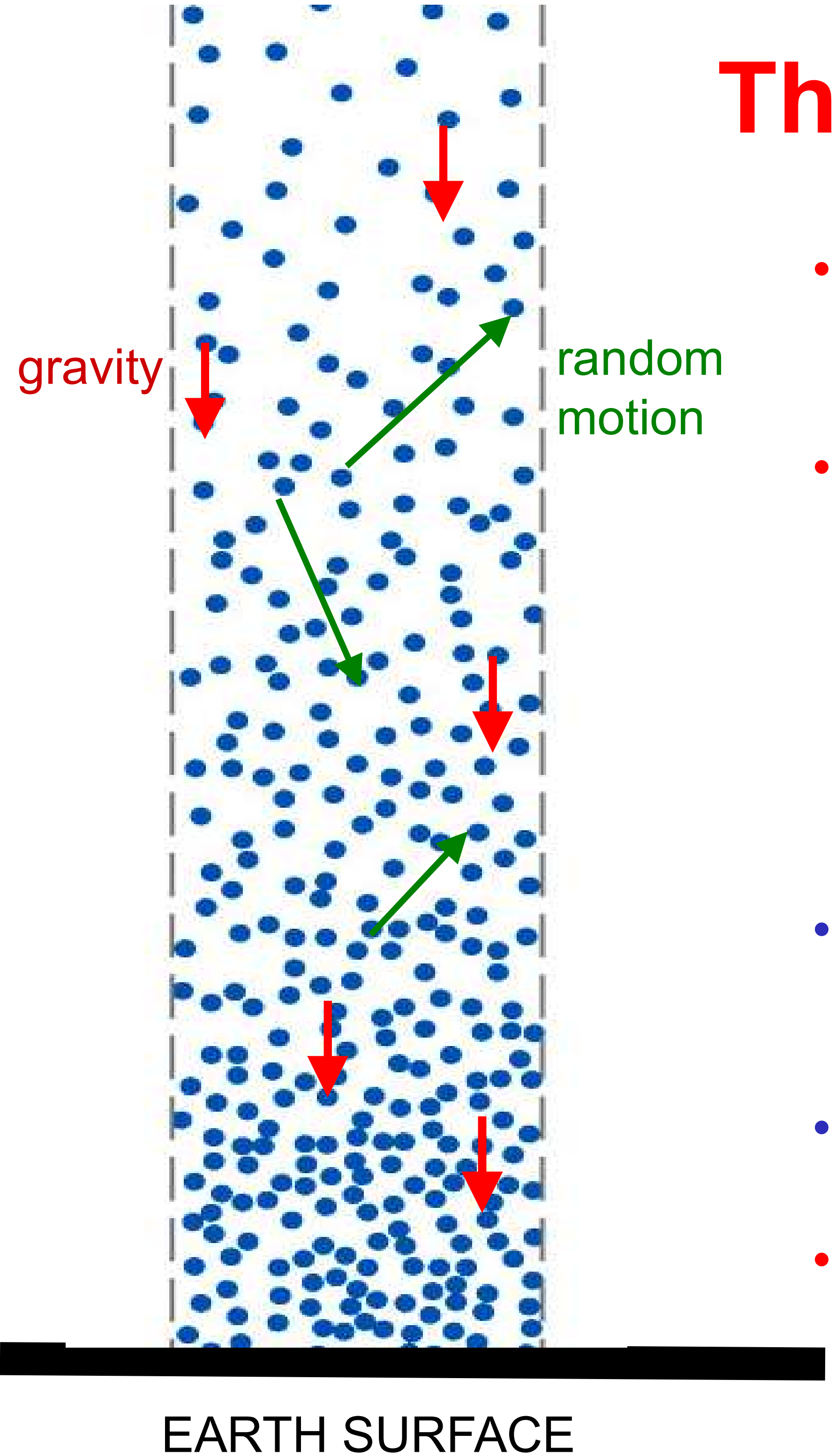


Atmospheric pressure and therefore the concentration of most molecules expressed in molecules / m^3 decrease exponentially with height. The x-axis is a log scale this makes exponential decrease look linear.

The relative proportion of major constituents to each other changes only in the thermosphere due to differences in the bond energies and mass.

Most water vapor remains contained in the troposphere due to poor mixing across the tropospause. The height profiles of photochemically formed species depend on radiation availability

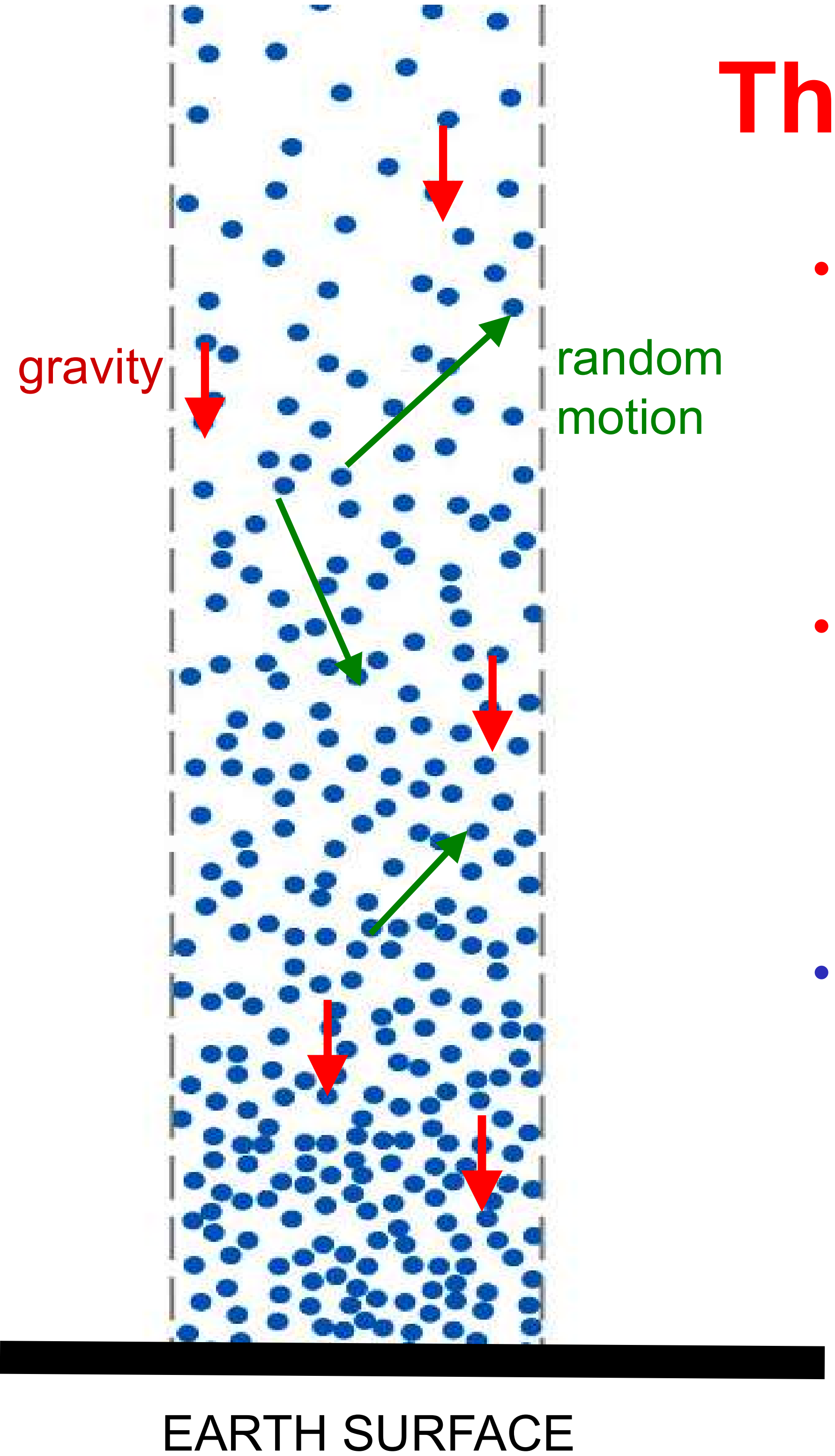
FIG. 7-1 Atmospheric vertical structure including temperature; composition and conventional names of atmospheric layers or altitude regions.



The Hydrostatic equation

- Atmospheric pressure is the weight of the overlying atmosphere per unit area of surface.
- SI unit for pressure is the Pascal (Pa): 1 Pa = 1 kg m^{-1} s^{-2} .
  Mean sea-level pressure:
  p = 1.013x10^5 Pa = 1013 hPa
  = 1013 mb
  = 1 atm
  = 760 mm Hg (torr)
- Weight of all air molecules is propagated to surface by random motion of molecules
- Random motion of molecules causes pressure to be applied in all directions
- Troposphere has 85% of atmospheric mass, stratosphere has 15%, Very little of the total mass is found above

The Hydrostatic equation



- Change of pressure with height is described by density and gravity

$$\frac{dp}{dz} = -\rho g$$

- For an ideal gas with mass m and molar mass M:

$$p = \frac{n}{V}RT = \frac{mRT}{MV} = \frac{\rho RT}{M}$$

- M can be defined for a gas mixture such as air or be used with m and M for individual gasses while computing partial pressures.

Molar mass of air assuming 78% N_2 and 21% O_2 most of rest Ar (M_{Ar}=40)
0.78x28 + 0.21 x 32 +40x0.01 = 21.84+ 6.72 + 0.06 =28.96 g/mol

Scale height

- Above 120 km there is very little turbulent mixing in the atmosphere the concentration of gases with respect to altitude starts changing with molecular mass.

$$\frac{dp}{dz} = -\rho g = -\frac{pMg}{RT}$$

- The relative proportion of major constituents wirth respect to each other changes with heavier inert gasses (Xe, Kr) enriched near the bottom of this layer and lighter ones (He) at the top.

$$\frac{dp_i}{p_i} = -\left(\frac{M_i g}{RT}\right) dz = -\frac{dz}{H_i}$$

- The scale height is defined as

$$H_i = \frac{RT}{M_i g}$$

and large for constituents with low M.

Lapse rate

•The rate of decrease of temperature with altitude (-dT/dz) is called the lapse rate. To determine whether an atmosphere is stable or unstable we need to compare its atmospheric lapse rate ($-dT_{ATM}/dz$) to the adiabatic lapse rate ($-dT_A/dz$)

•When an air parcel not in contact with ground behaves adiabatically its enthalpy H=E+pV is constant. If we define the geopotential at a given height as the work needed to move 1 unit of mass from sea level to that height

$$\Phi = \int_0^z g\,dz$$

•For one mole of a ideal gas with the constant pressure heat capacity Cp=29.09 [J mol^{-1} K^{-1}], M=28.96 [gmol^{-1}] and g= 9.81 [m s^{-2}] with J= kg m^2 s^{-2}

$$0 = dH = C_p dT + M d\Phi$$

$$C_p dT = -M d\Phi = -M g dz$$

•Dividing by dz and Cp we derive the dry adiabatic lapse rate:

$$\frac{dT}{dz} = -\frac{Mg}{C_p} = -\Gamma_d \qquad \Gamma_d = \frac{28.96[g\,mol^{-1}]9.81[m\,s^{-2}]}{29.09 \times 10^3[g\,m^2\,s^{-2}\,mol^{-1}\,K^{-1}]} \approx 9.8\left[\frac{K}{km}\right]$$

Potential temperature

•The potential temperature of an air parcel is defined as the temperature an air parcel would reach if brought adiabatically from its existing temperature and pressure to a standard pressure P0

$$\theta = T\left(\frac{P_0}{P}\right)^{R/C_p}$$

•In the absence of clouds and heat exchange potential temperature is constant for a parcel of air that moves through the atmosphere.

• A hot surface resulting in warm air at the bottom leads to thermal convection

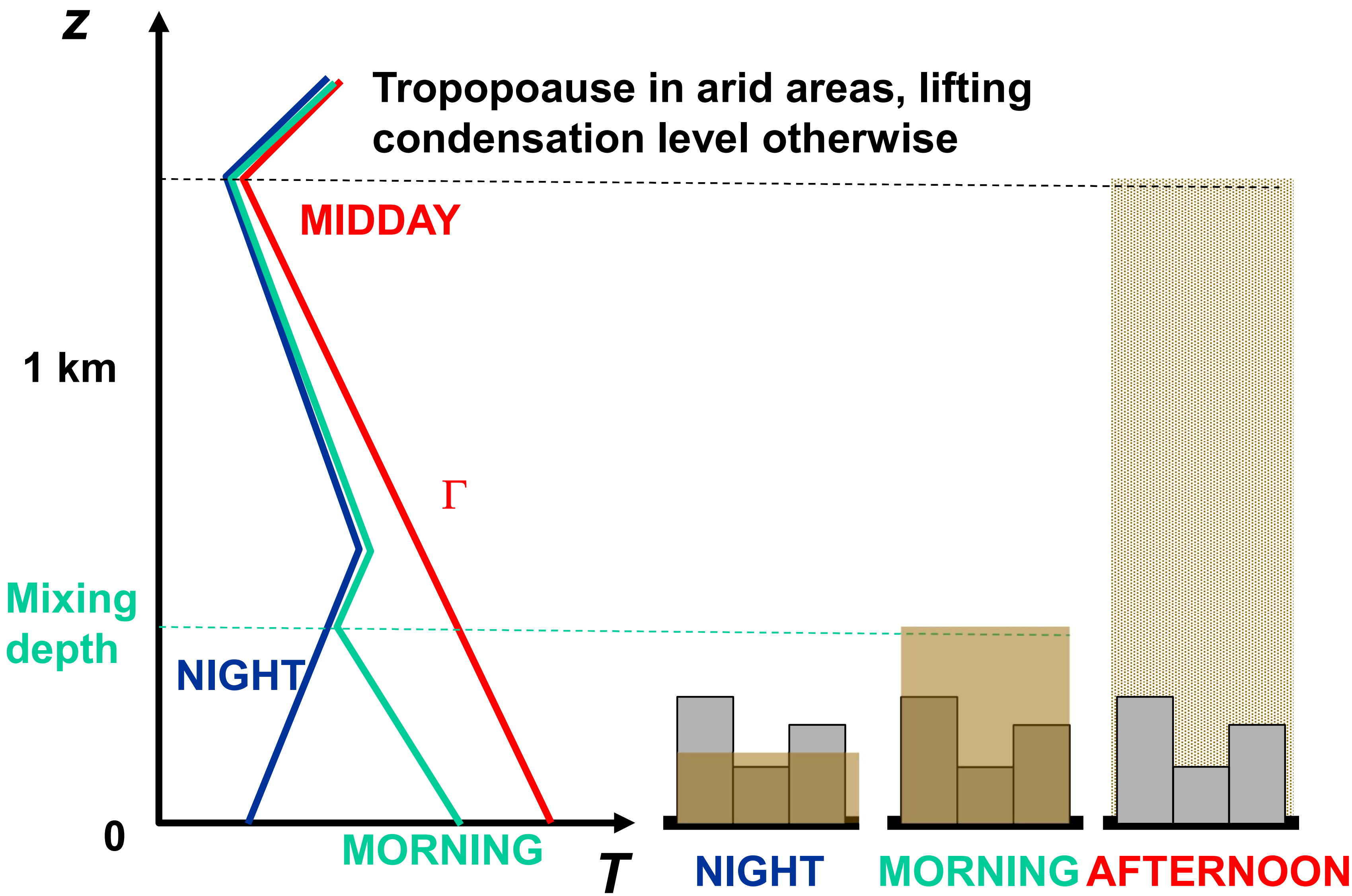
$$-\frac{dT_{ATM}}{dz} > \Gamma \qquad \text{unstable}$$

$$-\frac{dT_{ATM}}{dz} = \Gamma \qquad \text{neutral}$$

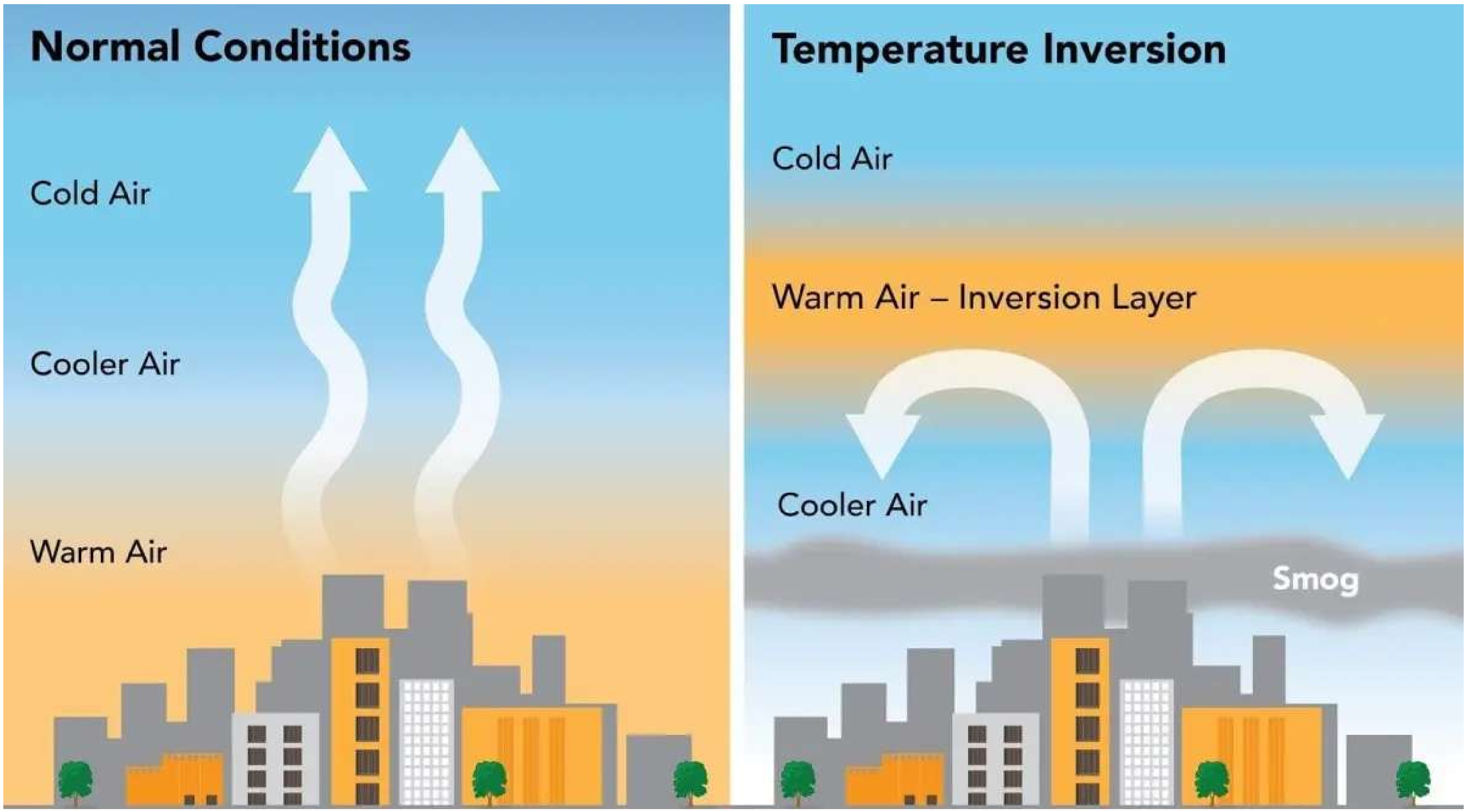
•Cold air below warm air ($dT_{ATM}/dz > 0$) leads to a stable atmosphere and an inversion

$$-\frac{dT_{ATM}}{dz} < \Gamma \qquad \text{stable}$$

With constant emissions atmospheric dilution has implications for concentrations.



Fatal air quality disasters typically happen in winter or at night.



Chemical constituents are well mixed in the PBL

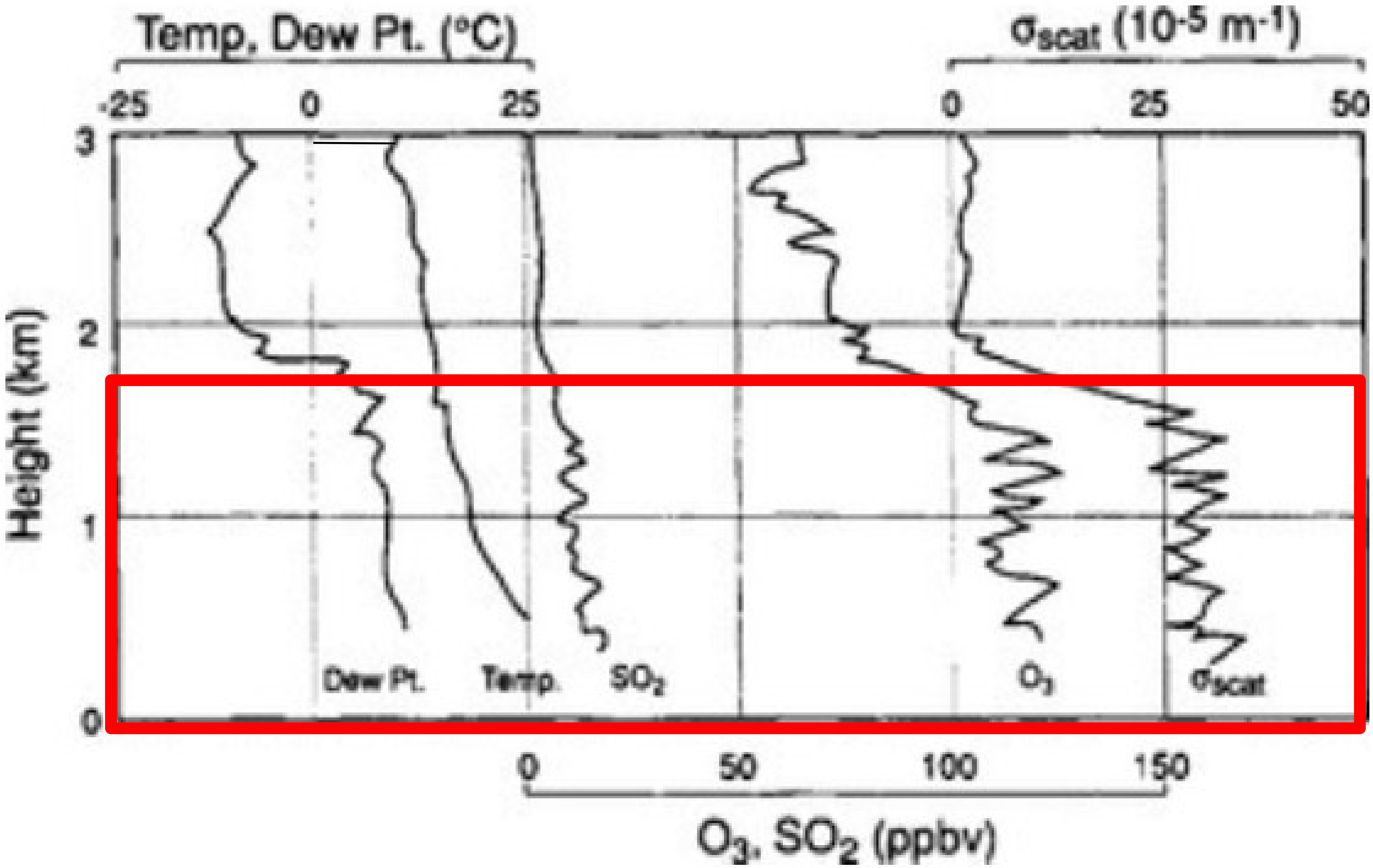
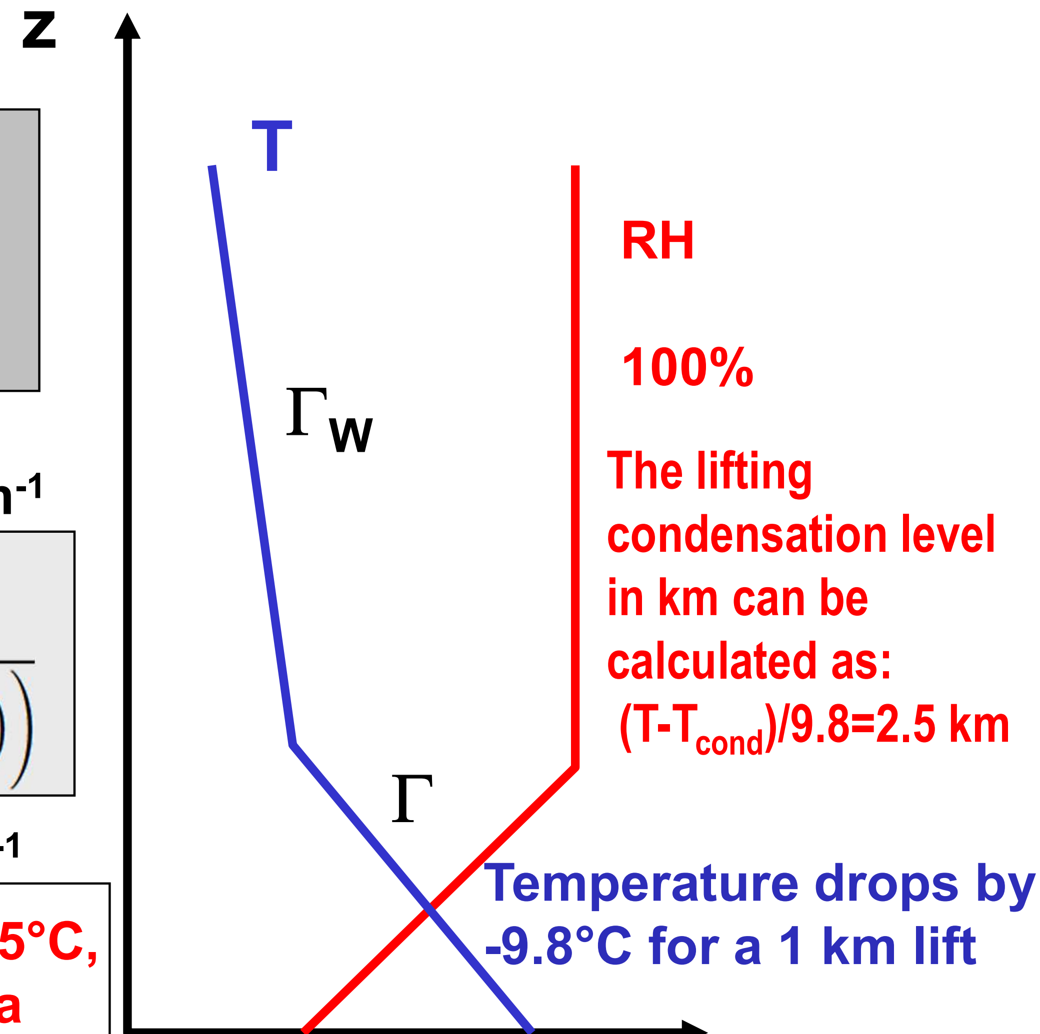


FIG. 7-2 Vertical profiles of physical (temperature, dew point, and backscatter coefficient) and chemical (ozone, sulfur dioxide) variables near Scranton, PA during the afternoon of 20 July 1978.

(Modified with permission from P. K. Mueller and G. M. Hidy (1982). "The Sulfate Regional Documentation of SURE Sampling Sites", EPRI report EA-1901, v. 3, Electric Power Research Institute.)

IN CLOUDY AIR PARCEL, HEAT RELEASE FROM H_2O CONDENSATION MODIFIES Γ

Wet adiabatic lapse rate Γ_W = 2-7 K km^{-1}



The updraft invigorates to m/s in cumulus clouds due to the latent heat release that increases buoyancy.

"Latent" heat release as H_2O condenses

Γ_W = **2-7 K km^{-1}**

@0.5°C
RH > 100%:
Cloud forms

$$T_{cond} = \frac{ln\left(\frac{p_{H_2O}}{0.61078}\right)237.3}{\left(17.27 - ln\left(\frac{p_{H_2O}}{0.61078}\right)\right)}$$

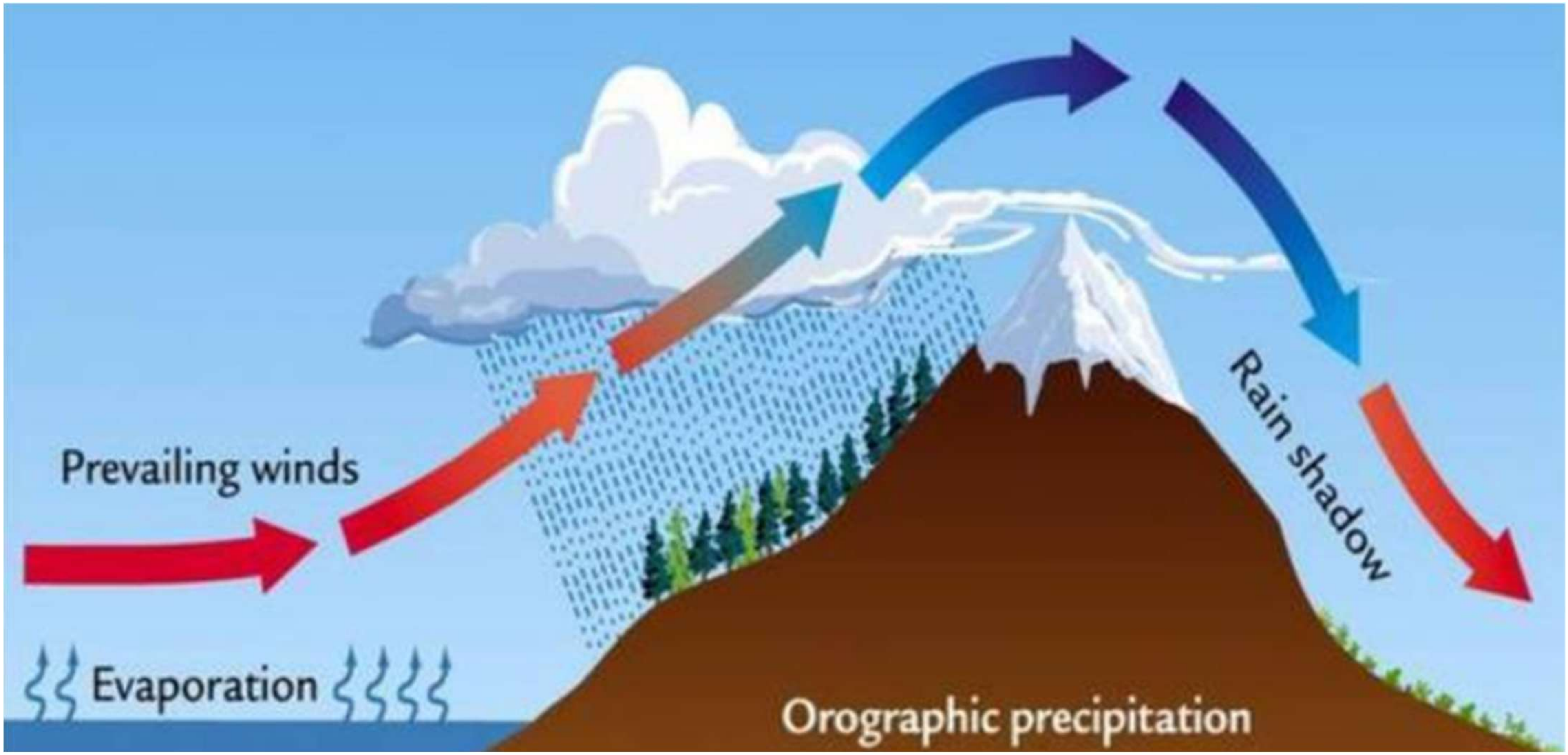
Γ = **9.8 K km^{-1}**

Starting with dry air 20% RH with a T of 25°C,
p_{H2Osat}=3.18 kPa & p_{H2O}=3.18*0.2=0.63 kPa

Exponential change in water vapor saturation pressure kPa for a change in temperature in °C (Thetens equation)

$$P = 0.61078\exp\left(\frac{17.27T}{T+237.3}\right)$$

You can see that lifting air always results in rain while descending air masses will always be dry



Water in the atmosphere

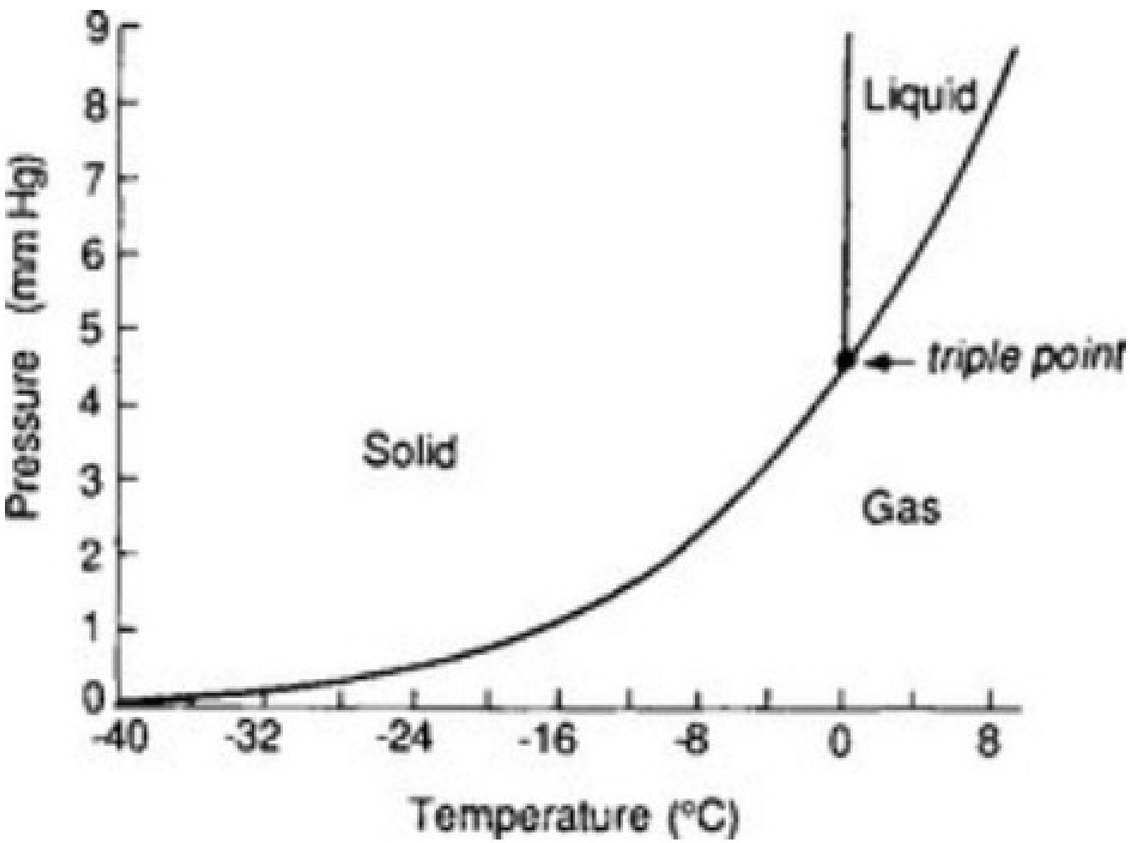
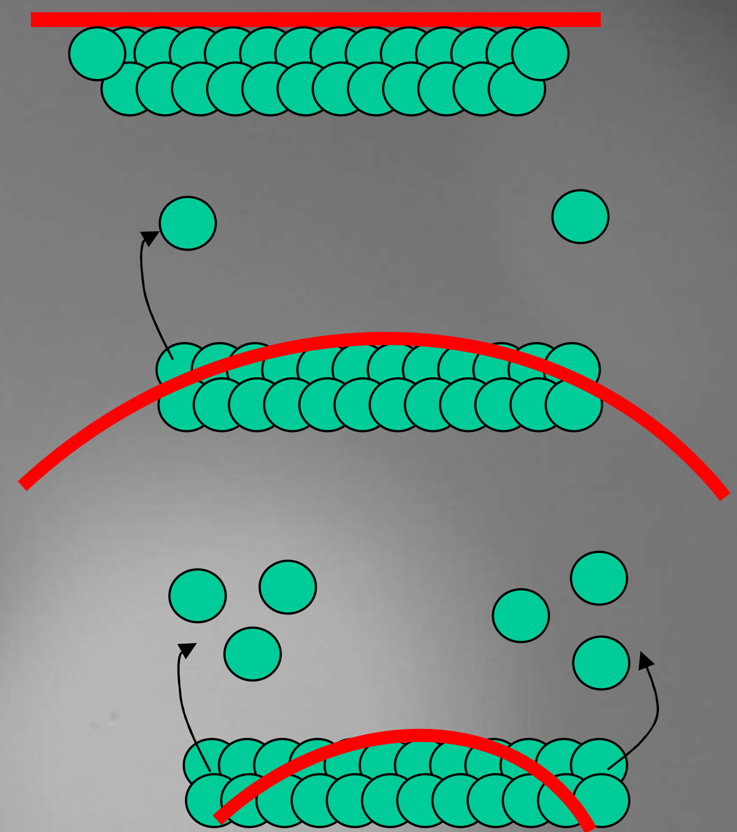


FIG. 7-9 *P–T* phase diagram for bulk water. (Based on data of the Smithsonian Meteorological Tables.)

- Vapor pressure very low at low temperature
- Can reach up to 25 g H_2O/m^3 in hot and humid weather
- Even in clouds relatively little of the total water is in the liquid phase 0.3-1 g H_2O/m^3
- Due to the high surface tension 300% RH would be required to nucleate pure water

Rain formation is delayed due to the Kelvin effect



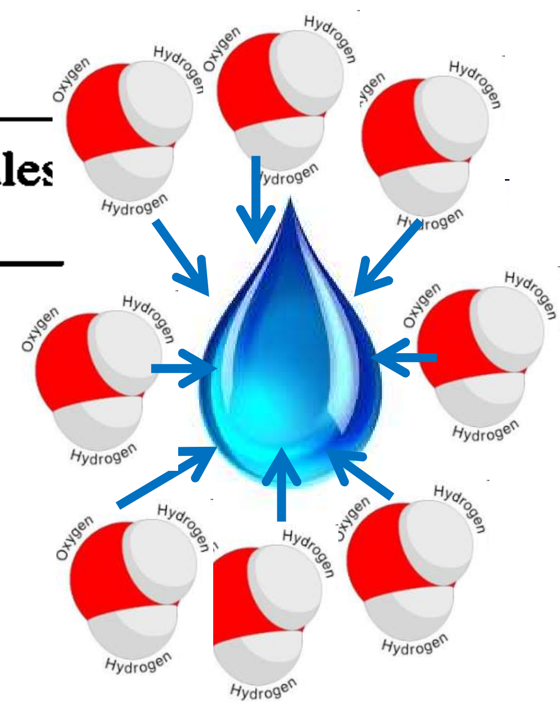
Equilibrium vapor pressure is higher over a curved surface than a flat one.
It depends on the curvature of the surface
Important for nucleation of new particles, lifetime of small droplets.

Kelvin Equation: $$S = \frac{p^{curved}}{p^{flat}} = \exp\left[\frac{2\sigma v_l}{k_B T R_p}\right] = \exp\left[\frac{2\sigma M}{\rho R T R_p}\right]$$

Relates molecular properties (molecular weight, surface tension, density) to the degree to which v.p. over curved surface is enhanced compared to that over a flat surface

TABLE 6.1. *Radii and Number of Molecules in Droplets of Pure Water in Equilibrium with the Vapor at 0°C*

Saturation ratio S	Critical radius r_c (μm)	Number of molecules n
1	∞	∞
1.01	1.208×10^{-1}	2.468×10^{8}
1.10	1.261×10^{-2}	2.807×10^{5}
1.5	2.964×10^{-3}	3.645×10^{3}
2	1.734×10^{-3}	730
3	1.094×10^{-3}	183
4	8.671×10^{-4}	91
5	7.468×10^{-4}	58
10	5.221×10^{-4}	20



Homogeneous nucleation of water vapor is impossible under atmospheric conditions. We need particles to serve as seeds for cloud droplets. Particles in the size range 50-100 nm

CONDENSATION CONSIDERING THE KELVIN EFFECT

$$P = 0.61078 \exp\left(\frac{17.27T}{T+237.3}\right)$$

$$S = \frac{p^{curved}}{p^{flat}} = \exp\left[\frac{2\sigma v_l}{k_B T R_p}\right] = \exp\left[\frac{2\sigma M}{\rho R T R_p}\right]$$

During monsoon measured temperature and RH in Mohali

~300 m AMSL, 980 hPa, T=37°C, RH=60

a) What is the absolute water vapor pressure?
3.76 kPa

b) At what height is S=1 based on the dry adiabatic lapse rate?
T_{cond}=28°C, 1.2 km AMSL

c) What will be S and Rp on which droplets form (assuming non-hygroscopic aerosol) for an air mass is lifted to 2000 m AMSL, T=26°C. S=3.76/3.36=1.12

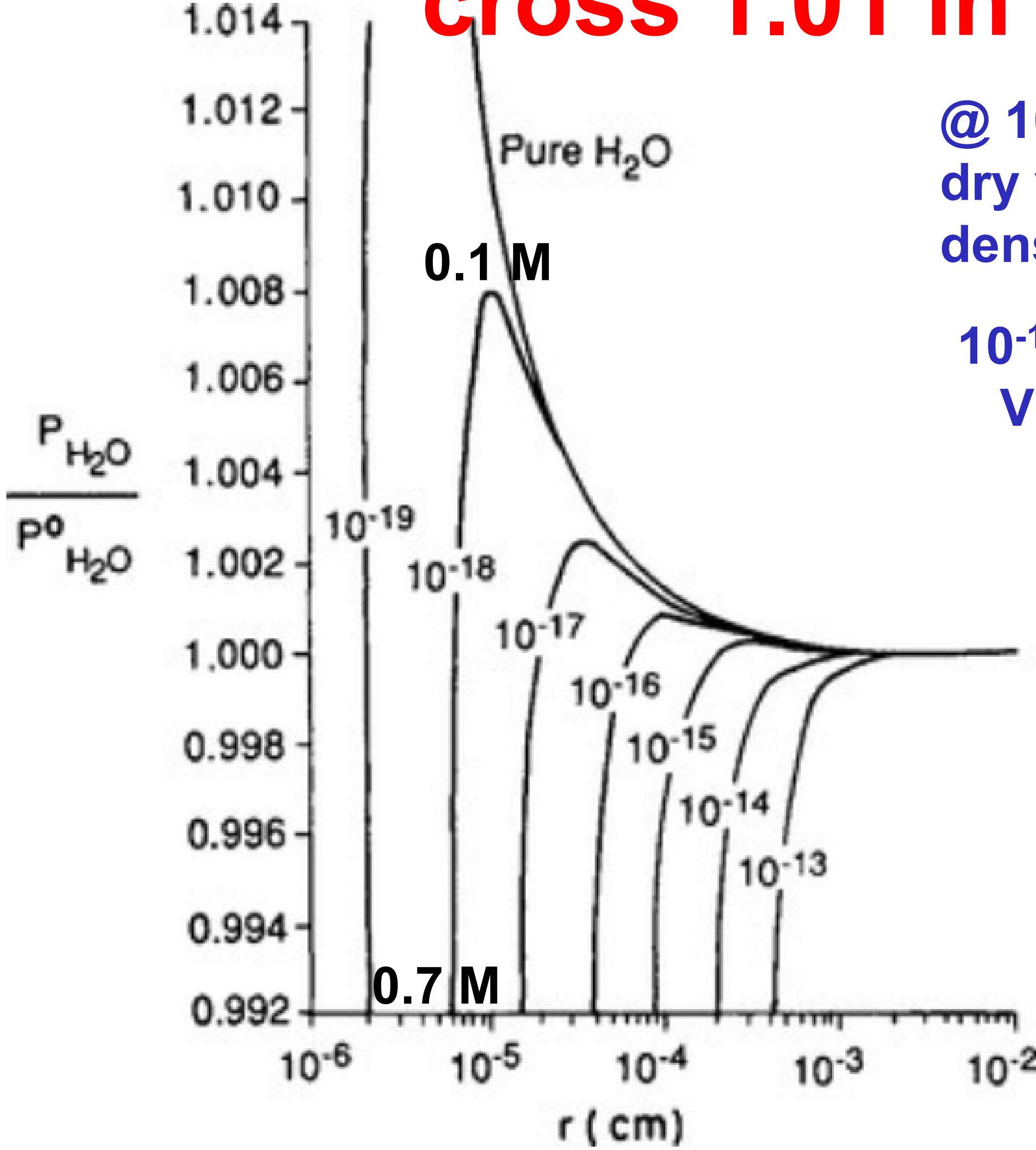
$$R_p = \frac{2 \times 18[\cancel{g}\ \cancel{mol^{-1}}] \times 73[\cancel{dyn}\ cm^{-1}]}{1[\cancel{g}\ cm^{-3}] \times 8.313 \times 10^7[\cancel{dyn}\ cm\ \cancel{mol^{-1}}\ \cancel{K^{-1}}] \times 299.15[\cancel{K}] \times ln(1.12)}$$
$$\approx 9.3 \times 10^{-7}[cm] \approx 9.3 \times 10^{-9} m \approx 9.3\ nm$$

d) Contrast the temperature of air with the temperature that the air mass would have assuming the transport represents adiabatic expansion to 2000 m without condensation. What is the wet adiabatic lapse rate in this case? ΔT should 7.8°C for 0.8 km is 2/0.8 km = 2.5°C /km

Latent heat of vaporization (0°C)	2.5×10^6 J/kg
Latent heat of freezing	3.3×10^5 J/kg
Ratio of density frozen/density liquid (0°C)	0.92
Surface tension	73 dyn/cm

as c R 0.08206 $m^3 \cdot atm/(kmole \cdot K)$ [8.314×10^7 $dyne \cdot cm/mole \cdot K$]

Water nucleation pure water and on salt Real CCN contain salt and S doesn't cross 1.01 in the real world



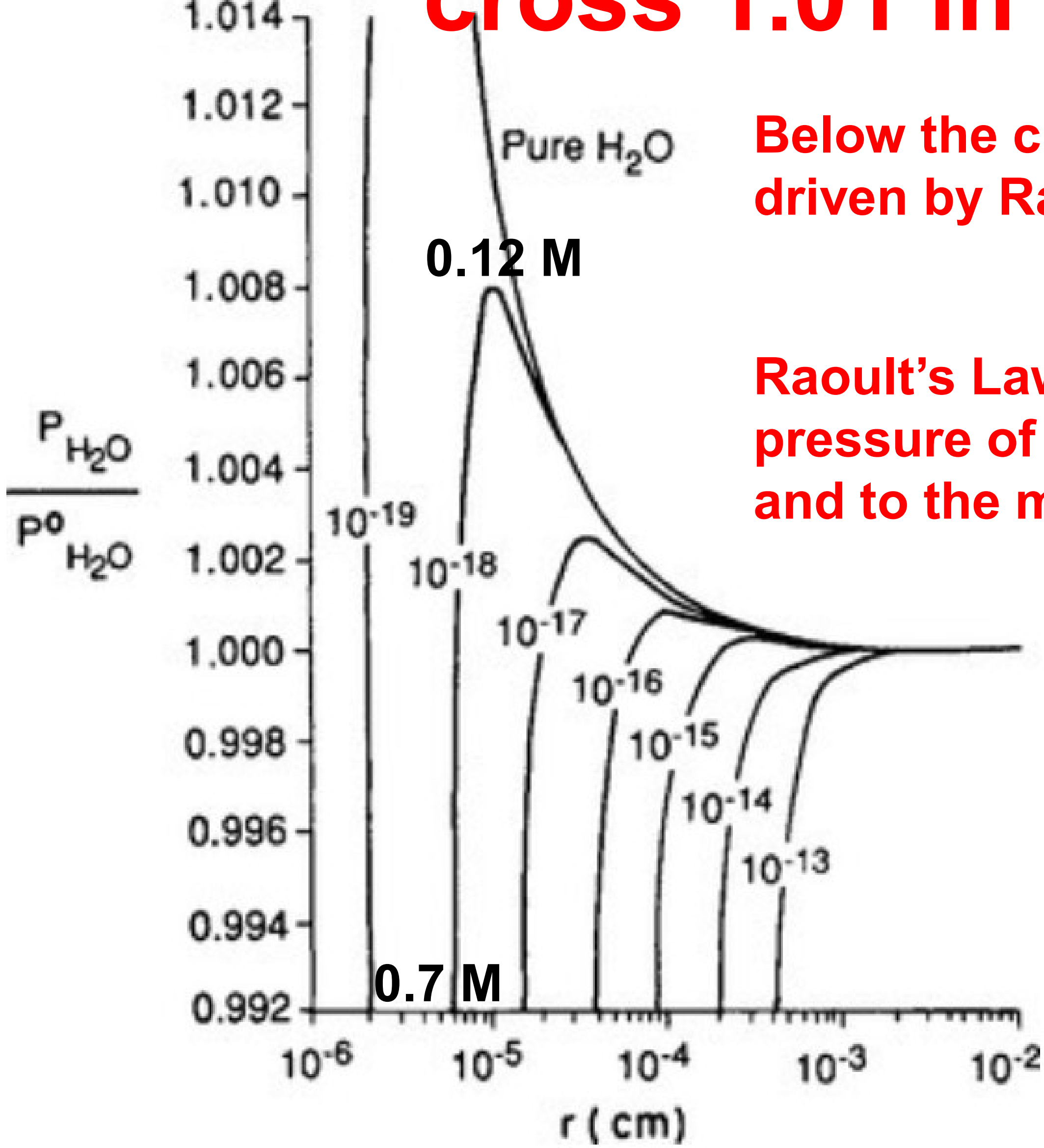
@ 10^{-18} mol= im/M what would be the dry volume of the NaCl particle? The density of NaCl is 2.2 g/cm^3

10^{-18} (23+35.5) /2=2.9*10^{-17}g (salt weight)
V=1.3*10^{-17} cm^3 => r_d=1.5*10^{-6} cm

@ 10^{-5} cm r what is Vs/Vw for 10^{-18} :
1.3*10^{-17}/4.2*10^{-15}=0.003
Most of the droplet is water there is ~0.1 mol/L NaCl

FIG. 7-10 KÖhler curves calculated for the saturation ratio $P_{H_2O} / P^0_{H_2O}$ of a water droplet as a function of droplet radius r. The quantity im/M is given as a parameter for each line, where m = mass of dissolved salt, M = molecular mass of the salt, i = number of ions created by each salt molecule in the droplet.

Water nucleation pure water and on salt Real CCN contain salt and S doesn't cross 1.01 in the real world



Below the critical radius hygroscopic growth is driven by Raoult's Law

$$e_{sol} = e_s \cdot (1 - \chi_s)$$

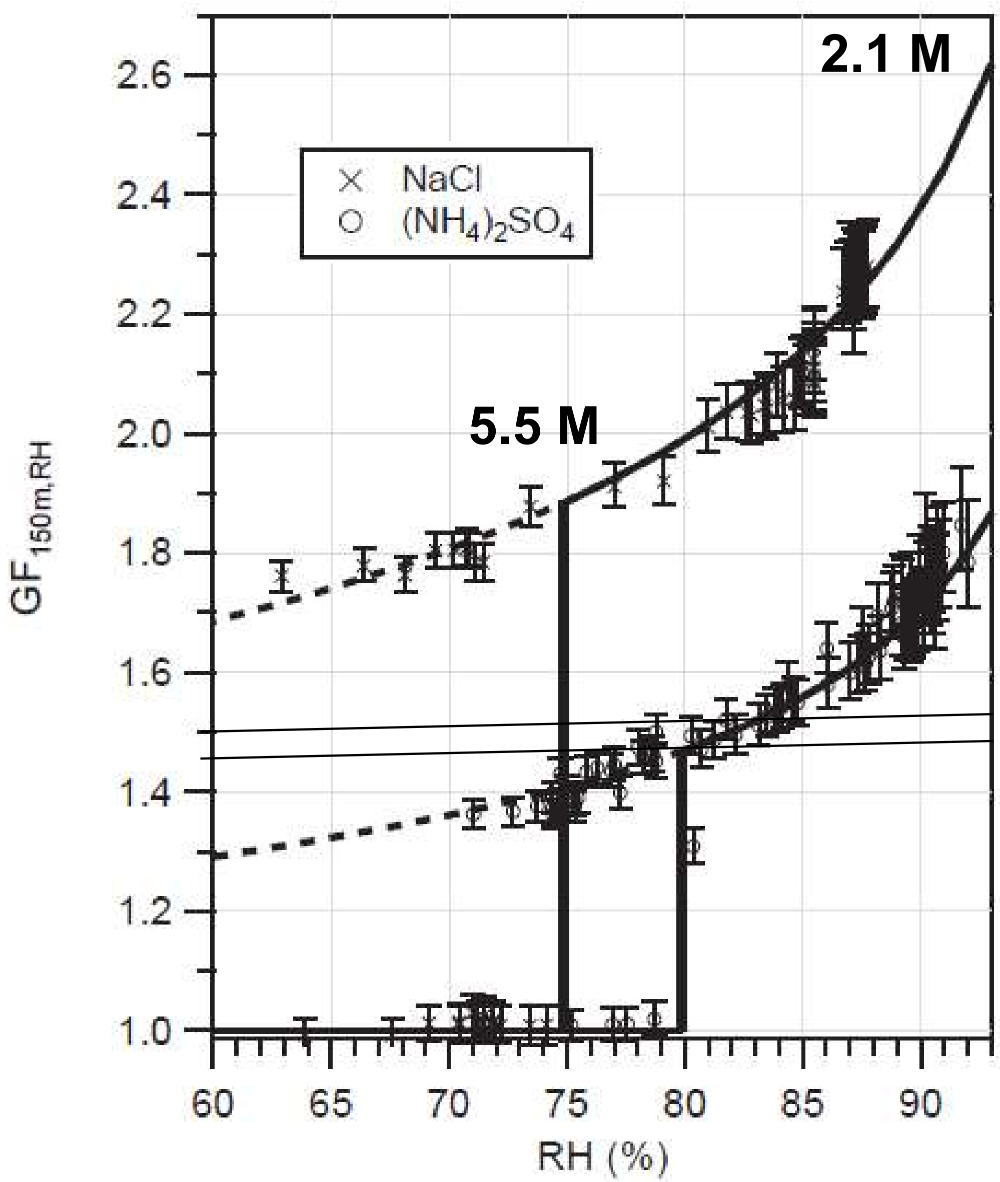
Raoult's Law relates the equilibrium vapor pressure of the solution (e_{sol}) to that of pure water and to the mole fraction

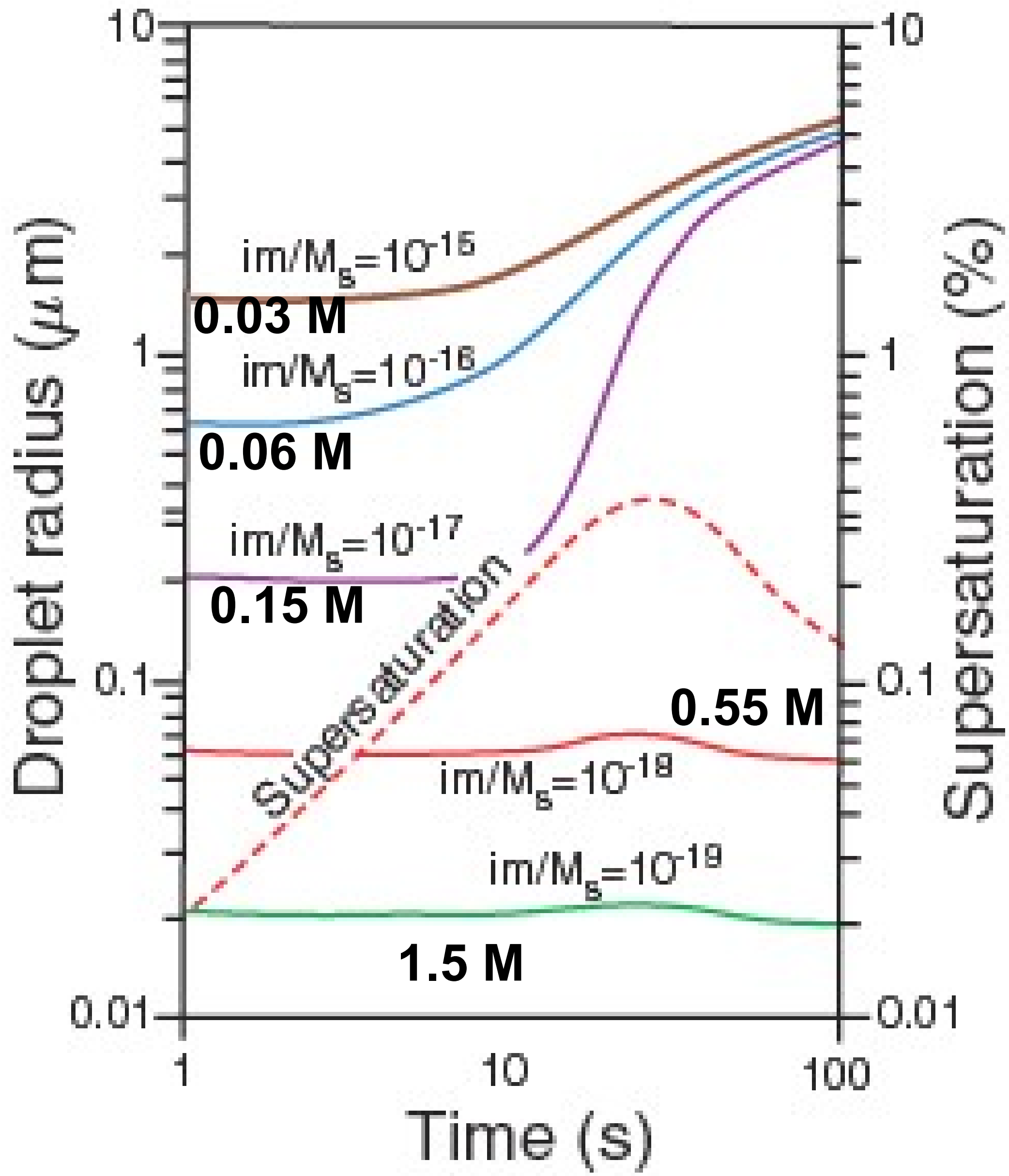
$$\chi_s \approx \frac{n_s}{n_L} = \frac{iN_s}{n_L V_{drop}} = \frac{iN_s}{n_L\left({4\pi r_d}^3/3\right)}$$

@10^{-5} cm r and 10^{-18} Xs=0.0043 and S= 1.008 instead of =1.012
@0.55*10^{-5} cm r and 10^{-18} Xs=0.026 and S= 0.992 instead of =1.019

$$S = \frac{p^{curved}}{p^{flat}} = \exp\left[\frac{2\sigma v_l}{k_B T R_p}\right] = \exp\left[\frac{2\sigma M}{\rho R T R_p}\right]$$

Salts grow early & rain drops become monodisperse fast





PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 15th September if you want to get them corrected before the Midsem exam

MCQ

1. The percentage of water vapor in the atmosphere is variable. It generally lies in wihch range
   a) 0.05-1%
   b) 1 to 3%
   c) 5% to 10%
   d) 10% to 20%
2. The atmosphere of the Earth is made up of _____ layers.
   a) 3
   b) 4
   c) 5
   d) 6
3. The ____________ contains the ozone layer.
   a) Troposphere
   b) Stratosphere
   c) Thermosphere
   d) Ionosphere
4. The troposphere contains ______ of the total atmospheric mass.
   a) 25
   b) 50
   c) 75
   d) 90

MCQ

1. The boundary between the troposphere and stratosphere is called
    a) Tropopause
    b) Stratopause
    c) Mesopause
    d) Thermopause
2. Which of the following is not considered an element of weather?
    a) Temperature
    b) Wind
    c) Sunshine
    d) Atmospheric pressure
3. Thermal inversion occurs when a layer of warm air forms over a layer of cool air. Which of the following are characteristics of thermal inversion?
    a) In high altitude areas, inversion lasts longer.
    b) Low-speed winds blow through the region.
    c) The region experiences dense ground fog.
    d) There is very dry air and no dew

Numerical

1) a) Calculate the water vapor pressure in an air parcel with a temperature of 25°C and 40% RH using the Thetens equation:

$$P = 0.61078 \exp\left(\frac{17.27T}{T + 237.3}\right)$$

b) Calculate at what temperature and lifting condensation level will 100% RH be reached if the air parcel is lifted

c) On the ascent from the lifting condensation level to 3 km the air only cools the 6°C. Calculate the wet adiabatic lapse rate of this air parcel.

d) The air parcel contains 100 nm sized hydrophobic soot . Use the Kelvin equation to determine at which S ratio the soot would start serving as CCN. The surface tension of water σ= 73 [dyn cm^{-1}] and R=8.314 x 10^{7} [dyn cm mol^{-1} K^{-1}] and water has the density 1 g cm^{-3}.

$$S = \frac{p^{curved}}{p^{flat}} = \exp\left[\frac{2\sigma v_l}{k_B T R_p}\right] = \exp\left[\frac{2\sigma M}{\rho R T R_p}\right]$$

Numerical

e) The air parcel also contains 20 nm sized $(NH_4)_2SO_4$. The density of this salt is 1.8 g cm^{-3}. Calculate how many g and mol of the salt are present in a spherical $(NH_4)_2SO_4$ particle of 20 nm diameter. V=1/6 pi d^3

f) This salt is prone to sub saturation hygroscopic growth and becomes a droplet with a wet radius that is 1.5 times the radius of the dry salt at 80% RH. What is the molarity of the solution initially formed?

g) Use the Kelvin equation to estimate the super saturation at which a particle of this wet diameter would normally grow and then use Raoult's Law to revise the estimate. Which particle will end up serving as cloud condensation nuclei, the 100 nm soot or the 20 nm sized $(NH_4)_2SO_4$

$$e_{sol} = e_s \cdot (1 - \chi_s) \qquad \chi_s \approx \frac{n_s}{n_L} = \frac{iN_s}{n_L V_{drop}} = \frac{iN_s}{n_L \left(4\pi r_d{}^3/3 \right)}$$

Numerical

1. Calculate the molar mass of air assuming 78% N_2 and 21% O_2 and 1% Ar (M_{Ar}=40)
2. Calculate the partial pressure of water in the Earth's atmosphere if the temperature is 20°C and the relative humidity is 50%
3. Calculate the relative humidity for the following atmospheric conditions 25°C, 1 atm, and a partial pressure of 0.0275 atm for water.
4. The airborne concentration of NO_2 on a day with 25°C, 1 atm, is measured to be 46 ppbv how many $\mu g/m^3$ are present.
5. The concentration of methane (CH4) has been determined as 9×10^{13} molecules / cm^3. Express this in ppmv

Long answer style

1. Draw the vertical structure of the atmosphere. Label the layers and explain how temperature and pressure change with altitude
2. Draw a vertical cross section of the troposphere from the equator to the pole which displays the motion of the major atmospheric circulation cells, the location of the two jet streams and the place were clouds will form.
3. Explain why water can not condense to form rain without aerosol.